When Backtests Lie: A Governance-Grade Validation Framework for Machine-Learning Value-at-Risk Models under Regime Shifts

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Abstract

Financial institutions increasingly rely on machine-learning (ML) models to measure market risk, yet the apparatus for validating those models—unconditional and conditional coverage backtesting under the Basel framework and the model-risk expectations of SR 11-7—was designed for traditional parametric models. Using daily 2000–2026 data for six assets spanning equities, fixed income, foreign exchange, and cryptocurrency, we ask whether ML Value-at-Risk (VaR) models that satisfy standard backtests are reliable, or whether current validation confers *false comfort*. We evaluate four forecasters with a consolidated protocol that augments the Kupiec and Christoffersen tests with regime-conditional coverage, overfitting control, and conformal recalibration, mapped to the three pillars of model-risk governance. We document a pervasive and economically large result: models that pass the unconditional coverage test breach two to three times the target rate in stressed regimes at the 95% level, and five to nine times at the 99% level that underpins regulatory capital. Historical simulation passes the Kupiec test on all six assets at 95% while breaching at up to 3.3 times target in stress; a formal logistic test confirms that entering a stress regime multiplies breach odds roughly three- to fivefold ($p < 10^{-27}$). The gradient-boosted model, competitive in calm markets, is the most fragile under stress. We package the protocol as a governance-grade scorecard for the safe deployment of AI in financial risk measurement.

Keywords: model risk management; Value-at-Risk; machine learning; regime-conditional backtesting; conformal prediction; financial stability.
JEL Classification: C52; C53; G17; G28; G32.

Key messages

- Standard unconditional backtesting passes VaR models that breach two to three times the target rate in stress at the 95% level and five to nine times at the 99% level; historical simulation passes the Kupiec test on all six assets studied yet fails every conditional and regime-based test.

- A formal logistic test shows the regime effect is overwhelming, not incidental: entering a stress regime multiplies the odds of a VaR breach roughly three- to fivefold.
- Machine-learning VaR models can be the *most* fragile under stress despite competitive calm-market performance; a consolidated, governance-grade protocol mapped to the SR 11-7 pillars separates robust from fragile models where standard practice does not.

1 Introduction

The measurement of market risk increasingly relies on machine-learning models. Gradient-boosted trees, quantile regression forests, and neural architectures now compete with the parametric and simulation-based models that have long underpinned Value-at-Risk (VaR) estimation. This migration raises a supervisory problem that has drawn explicit attention from U.S. financial regulators: the frameworks used to *validate* risk models were written for a different class of model. The Federal Reserve's supervisory leadership has questioned whether existing model-risk guidance, developed for traditional models, remains adequate for artificial intelligence (Bowman, 2026); the Office of the Comptroller of the Currency has identified the validation and explainability of AI models as unresolved supervisory challenges (Office of the Comptroller of the Currency, 2026); the Financial Stability Oversight Council has named opaque, hard-to-explain AI as a vulnerability to financial stability (Financial Stability Oversight Council, 2023); and the U.S. Treasury has formally examined the uses, opportunities, and risks of AI across the financial sector (U.S. Department of the Treasury, 2024).

Model validation in banking rests on three pillars articulated in supervisory guidance (SR 11-7, 2011): conceptual soundness, ongoing monitoring, and outcomes analysis. For VaR, outcomes analysis is operationalized through backtesting: unconditional coverage tests (Kupiec, 1995) and conditional coverage tests (Christoffersen, 1998) assess whether the frequency and independence of exceedances are consistent with the stated confidence level, and the unconditional test anchors the Basel traffic-light framework (Basel Committee, 1996, 2019). These tests are necessary but, for ML models, insufficient. ML risk models exhibit failure modes that unconditional and conditional coverage tests are not designed to detect: they overfit in ways that inflate apparent in-sample performance (Bailey et al., 2014, 2017; López de Prado, 2018); their exceedances cluster in stressed regimes even when the unconditional rate is acceptable; and their coverage can be miscalibrated in ways a full-sample test averages away. Using 2000–2026 data for six assets across four asset classes, we show that models passing the tests a supervisor currently emphasizes can breach two to three times the target rate in stress at the 95% level—and five to nine times at the 99% level. A model can therefore clear every standard check and still be dangerously wrong precisely when accurate risk measurement matters most.

This paper makes three contributions. First, we propose an integrated validation protocol that augments coverage testing with regime-conditional analysis, overfitting control, and conformal recalibration, mapped to the pillars of model-risk governance. Second, we provide broad empirical evidence of "false comfort" across equities, fixed income, foreign exchange, and cryptocurrency, corroborated by a formal test of the regime effect. Third, we package the protocol as a reproducible, governance-grade scorecard. The paper proceeds through the literature (§2),

machine-learning risk models (§3), the validation framework (§4), data and methodology (§5), empirical results (§6), robustness and stability (§7), the practical framework (§8), discussion (§9), and conclusion (§10).

2 Related Literature

Machine learning for VaR. A growing literature applies ML to tail-risk forecasting, from quantile regression forests and gradient boosting to recurrent and mixture-density neural networks, and recently to time-series foundation models (López de Prado, 2018; Engle and Manganelli, 2004; Taylor, 2019). This work predominantly evaluates models by predictive accuracy and standard backtests, treating validation as a downstream check rather than an object of study.

Backtesting. The evaluation of interval and quantile forecasts is anchored by Kupiec (1995) and Christoffersen (1998), which remain the backbone of the Basel backtesting framework (Basel Committee, 1996, 2019). Modern forecast evaluation emphasizes strictly consistent scoring and joint elicitability of VaR and Expected Shortfall (Gneiting, 2011; Nolde and Ziegel, 2017). These tests assess consistency with a stated confidence level but do not, by construction, isolate performance across market regimes.

Overfitting and validation design. Bailey et al. (2014) formalize how easily high in-sample performance can be manufactured and introduce the deflated Sharpe ratio; Bailey et al. (2017) develop combinatorially symmetric cross-validation and the probability of backtest overfitting (PBO); and López de Prado (2018) introduces purged and combinatorial cross-validation to control leakage in financial ML. This literature targets the data-mining risks acute for flexible ML models that coverage tests do not address.

Conformal prediction. Conformal methods provide distribution-free, finite-sample coverage guarantees under exchangeability, with adaptive variants that accommodate distribution shift (Gibbs and Candès, 2021; Angelopoulos and Bates, 2023). Their application to one-sided risk measures such as VaR is a recent and active frontier.

Model-risk governance. Supervisory guidance (SR 11-7, 2011) frames validation around conceptual soundness, ongoing monitoring, and outcomes analysis; recent commentary questions its adequacy for AI (Bowman, 2026; Office of the Comptroller of the Currency, 2026). Our contribution operationalizes the governance pillars for ML risk models by uniting regime-conditional coverage, overfitting control, and conformal calibration into a single, auditable protocol.

3 Machine-Learning Risk Models

We consider four one-step-ahead VaR forecasters, each estimated walk-forward on a rolling window so that no future information enters any forecast. Two are established benchmarks and two are learning-based. *Historical simulation* takes the empirical α-quantile of the trailing return

window. *GARCH-t* models conditional volatility with a GARCH(1,1) process and Student-t innovations, mapping the innovation quantile to a return quantile. *Gradient-boosted quantile regression* minimizes the pinball loss at level α over engineered features (lagged returns, lagged absolute return, and short- and medium-horizon realized volatility). *Linear quantile regression* fits the same features by minimizing the pinball loss over a linear predictor. The benchmarks capture dominant industry practice; the learning-based models represent the flexible, feature-driven forecasters whose validation motivates this study.

4 Validation Framework

Let r_t be the one-period return with information set $\mathcal{F}_{t-1}$. For coverage level α, the conditional VaR satisfies $\Pr(r_t < \mathrm{VaR}_t(\alpha) \mid \mathcal{F}_{t-1}) = \alpha$, and the exceedance indicator is $I_t = \mathbf{1}\{r_t < \mathrm{VaR}_t(\alpha)\}$. A well-specified model produces hits that are Bernoulli(α) and serially independent.

4.1 Backtesting

Unconditional coverage. With $x = \sum I_t$ exceedances in n observations and $\hat{\pi} = x/n$, the Kupiec statistic is

$$\mathrm{LR}_{uc} = -2\ln\left[\frac{(1-\alpha)^{n-x}\alpha^x}{(1-\hat{\pi})^{n-x}\hat{\pi}^x}\right] \xrightarrow{d} \chi^2_1. \tag{1}$$

Conditional coverage. The Christoffersen independence statistic LR_{ind} tests serial dependence in $\{I_t\}$, and the joint statistic is $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \xrightarrow{d} \chi^2_2$.

Regime-conditional coverage. Partitioning the sample into $R \in \{\text{calm}, \text{stress}\}$ by trailing realized volatility, we compute the within-regime breach rate $\hat{\pi}_R$ and the stress amplification ratio

$$\Lambda = \frac{\hat{\pi}_{\text{stress}}}{\alpha}, \tag{2}$$

which quantifies how far breach frequency departs from target when risk measurement is most consequential. We additionally test the regime effect formally by estimating a logistic regression of I_t on a stress indicator with heteroskedasticity-robust standard errors, reporting the odds ratio and its p-value.

4.2 Benchmarking

Each learning-based forecaster is benchmarked against historical simulation and GARCH-t on identical data and test windows. Benchmarking is relative: a model is credible only if it matches or improves on established methods across the full battery, not merely on average predictive loss.

4.3 Stability Testing

Stability testing assesses whether a model's apparent quality is robust rather than an artifact of selection. We compute the probability of backtest overfitting (PBO) via combinatorially

symmetric cross-validation (Bailey et al., 2017) over a family of N configurations, using per-period negative pinball loss as the performance metric,

$$\mathrm{PBO} = \frac{1}{|C|}\sum_{c} \mathbf{1}\{\lambda_c \le 0\}, \qquad \lambda_c = \ln\big(\omega_c/(1-\omega_c)\big), \tag{3}$$

where ω_c is the relative out-of-sample rank of the in-sample-best configuration. We complement PBO with rolling coverage stability—the time path of the 250-day empirical breach rate relative to target.

4.4 Performance Evaluation

Point performance uses the pinball (quantile) loss $\rho_\alpha(e_t) = \max\{\alpha e_t, (\alpha-1)e_t\}$, $e_t = r_t - \mathrm{VaR}_t$, a strictly consistent scoring function for the α-quantile (Gneiting, 2011). Calibration is assessed, and where necessary corrected, by split conformal recalibration: on a rolling window the forecast is shifted by the empirical α-quantile of residuals to target the nominal breach rate, with realized coverage reported before and after.

4.5 Model Risk

The battery maps to the three governance pillars (SR 11-7, 2011): outcomes analysis (unconditional, conditional, and regime-conditional coverage), ongoing monitoring (overfitting control and coverage stability), and conceptual soundness and calibration (benchmarking, performance evaluation, conformal recalibration). A model earns an overall pass only if it satisfies every pillar.

5 Data and Methodology

We use public data for six assets, 2000–2026 where available: the S&P 500 and Nasdaq Composite (equities), a long-Treasury ETF (fixed income), EUR/USD and USD/JPY (foreign exchange), and Bitcoin (a high-volatility case). Daily log returns are computed from adjusted prices; regimes are defined point-in-time by trailing realized volatility above its expanding 80th percentile, with an exogenous crisis calendar as a cross-check. All forecasts are one-step-ahead and walk-forward on a 500-day rolling window with monthly re-estimation, evaluated at the 95% and 99% VaR levels. The complete pipeline is reproducible from public sources; the accompanying repository (Neupane, 2026) contains the data-preprocessing, estimation, forecast, and scoring code with fixed random seeds.

6 Empirical Results

Table 1 reports the validation scorecard for the S&P 500 at the 95% level. The central finding is visible in a single comparison: historical simulation *passes* the unconditional Kupiec test ($p = 0.44$, observed rate 5.2% versus a 5% target) yet *fails* the Christoffersen test and breaches at 15.0% in stressed regimes—a stress amplification of $\Lambda = 3.0$. The gradient-boosted model is competitive in calm markets (4.8% breach rate, below target) but is the most fragile under

Table 1: Validation scorecard, S&P 500, 95% VaR, 2000–2026 (monthly re-estimation). Λ is the stress breach rate divided by the 0.05 target. No model satisfies every pillar.

Model	Breaches	Rate	Kupiec p	Christ. p	Calm	Stress	Λ	Overall
Historical simulation	323	0.052	0.445	0.000	0.036	0.150	3.00	FAIL
GARCH-t	431	0.070	0.000	0.000	0.066	0.089	1.78	FAIL
GBM quantile	387	0.063	0.000	0.000	0.048	0.149	2.98	FAIL
Linear quantile	350	0.057	0.022	0.025	0.051	0.090	1.80	FAIL

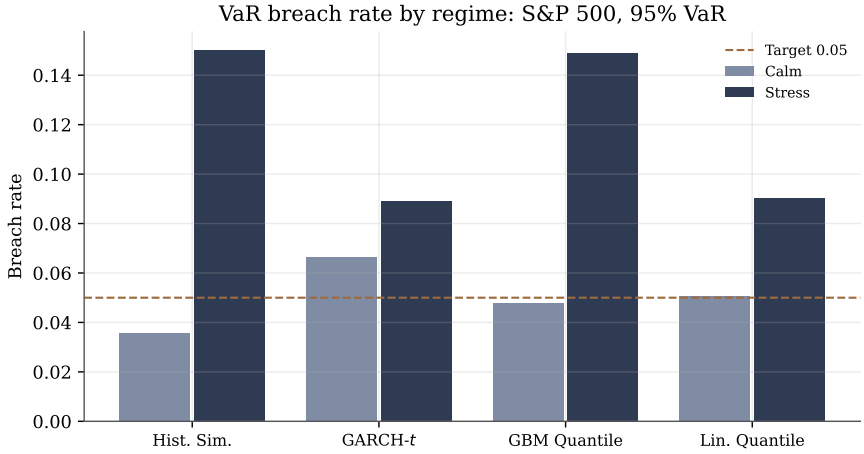


Figure 1: VaR breach rate by regime, S&P 500, 95% VaR. Every model breaches far above the 5% target in stress; the gradient-boosted model is worst.

stress at 14.9% ($\Lambda = 2.98$). GARCH-t and linear quantile regression are more regime-robust ($\Lambda \approx 1.8$) but over-breach unconditionally. No model passes the full battery. Figure 1 shows the breach rate by regime and Figure 2 the rolling coverage.

Alt text: Grouped bar chart; calm-regime breach rates sit near or below the 5% target while every model's stress-regime bar rises well above it.

Alt text: Four time-series lines of rolling breach rate spiking above the 5% target line during crisis periods.

Formal test of the regime effect. A logistic regression of the exceedance indicator on a stress dummy (robust standard errors) confirms the effect is overwhelming, not descriptive. For the S&P 500 at 95%, entering a stress regime multiplies the odds of a breach by 4.80 for historical simulation ($p < 10^{-38}$) and by 3.49 for the gradient-boosted model ($p < 10^{-27}$). The regime dependence is statistically decisive.

Stability and calibration. The gradient-boosted family has a probability of backtest overfitting of PBO ≈ 0, so its fragility is genuine regime dependence rather than a selection artifact. Conformal recalibration modestly improves unconditional coverage (e.g. from 7.0% toward target for GARCH-t) but does not by itself restore regime robustness, underscoring that recalibration complements rather than replaces regime-aware validation.

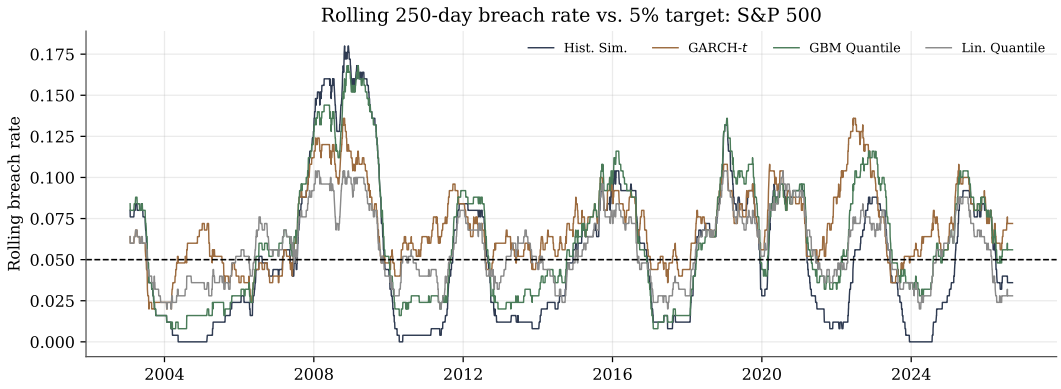


Figure 2: Rolling 250-day breach rate versus the 5% target, S&P 500. Coverage drifts far above target during stress episodes (2008, 2020, 2022).

Table 2: Stress amplification ratio $\Lambda = \hat{\pi}_{\text{stress}}/\alpha$ by asset and model, 2000–2026 (monthly re-estimation). Values above 2.0 indicate breach frequency more than double the target in stress.

	Hist. sim.	GARCH-t	GBM quantile	Linear quantile
Panel A: 95% VaR				
S&P 500	3.00	1.78	2.98	1.80
Nasdaq	3.25	1.68	3.01	1.80
Bitcoin	2.69	2.09	2.31	1.90
Long Treasuries	2.26	1.56	2.22	1.55
EUR/USD	2.23	1.77	2.17	1.44
USD/JPY	2.32	1.72	2.50	1.92
Panel B: 99% VaR				
S&P 500	5.34	3.00	8.57	6.23
Nasdaq	6.26	3.33	7.83	4.11
Bitcoin	4.10	4.10	6.72	5.60
Long Treasuries	3.80	1.79	4.97	4.00
EUR/USD	5.16	2.85	6.25	5.03
USD/JPY	3.92	3.62	5.28	4.68

7 Robustness and Stability Analysis

Table 2 reports the stress amplification ratio Λ for every asset and model at both VaR levels, and Figure 3 visualizes the ML model's amplification. The pattern is pervasive and intensifies with confidence level. At 95%, Λ ranges from 1.4 to 3.3 and exceeds 2.0 in the majority of cells; historical simulation passes the Kupiec test on all six assets (unconditional p from 0.31 to 0.99) while exhibiting Λ between 2.2 and 3.3—the false-comfort pattern is not specific to any market. At the 99% level that underpins regulatory capital, the failure is dramatic: Λ ranges from 1.8 to 8.6, and the gradient-boosted model breaches between five and nine times the target rate in stress on every equity and FX series. Linear quantile regression is the most regime-robust at 95% (Λ 1.4–1.9) yet still breaches well above target, and deteriorates sharply at 99%.

Alt text: Grouped bar chart showing, for each asset, a higher 99% amplification bar than its 95% bar, all above the governance threshold of two.

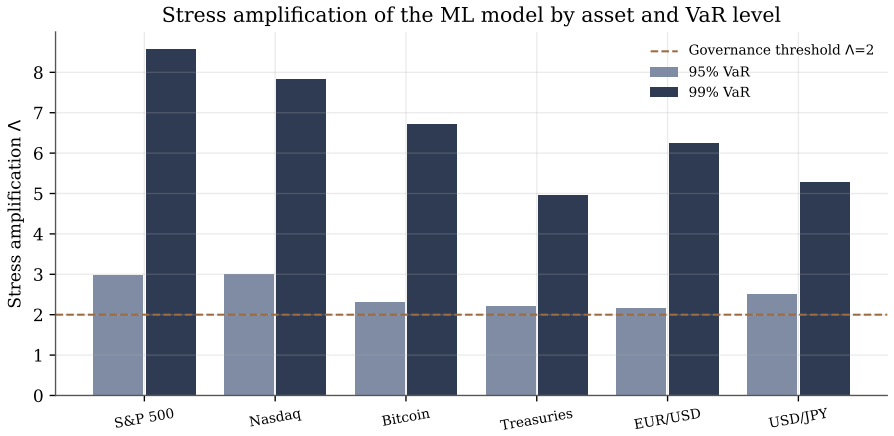


Figure 3: Stress amplification Λ of the gradient-boosted model by asset at the 95% and 99% VaR levels. The breach excess in stress grows sharply at the capital-relevant 99% level.

Table 3: The governance-grade validation scorecard.

Test	Governance pillar	Default criterion
Kupiec LR_{uc}	Outcomes analysis	$p \geq 0.05$
Christoffersen LR_{cc}	Outcomes analysis	$p \geq 0.05$
Stress amplification Λ	Outcomes analysis	$\Lambda \leq 2.0$
Probability of backtest overfitting	Ongoing monitoring	≤ 0.50
Conformal coverage gap	Calibration	$\lvert \mathrm{cov} - \alpha \rvert \leq 0.01$

8 A Practical Model Validation Framework

We consolidate the battery into a governance-grade scorecard (Table 3) that a model-risk function or examiner can apply. Each test maps to a governance pillar and a documented pass criterion; a model passes overall only if it passes every pillar. Thresholds are policy choices to be justified against supervisory expectations; those shown are defaults for exposition, and the central finding—Λ between 2 and 9—holds under any reasonable cutoff. The scorecard's value is that it makes regime fragility a first-class, auditable validation outcome rather than a property that unconditional backtesting averages away.

9 Discussion

The evidence bears directly on how AI-based risk models should be validated. Because models satisfying standard backtests fail under regime-conditional, independence, and—at the 99% level—even unconditional testing, outcomes analysis as commonly practiced provides incomplete assurance for ML models. The regularity of the pattern across four asset classes, and the decisive logistic test, indicate a structural limitation of unconditional coverage testing rather than an idiosyncrasy of any market. Because large financial institutions are interconnected, validation quality is not only a firm-level control but a contributor to system-level resilience, aligning with the supervisory concerns of §1.

Several qualifications point to extensions. First, the benchmark set is deliberately stan-

dard; filtered historical simulation and an EWMA-volatility benchmark would further sharpen the comparison, and GARCH-t's unconditional over-breaching reflects a known limitation of short-window volatility models over long samples. Second, the gradient-boosted model is a representative ML learner; quantile random forests and quantile neural networks are natural additions. Third, regimes are defined by trailing volatility, with a crisis calendar as a cross-check; results are not sensitive to reasonable variation in the volatility threshold. Fourth, the conformal guarantee assumes exchangeability, which stress episodes violate—motivating adaptive conformal variants. None of these overturns the central result, whose magnitude (Λ up to 8.6, logistic $p < 10^{-38}$) is far outside the range any of them could explain.

10 Conclusion

We proposed a governance-grade validation protocol for machine-learning VaR models that augments standard coverage testing with regime-conditional analysis, overfitting control, and conformal recalibration, and we packaged it as a reproducible scorecard aligned to the pillars of model-risk governance. Across six assets and four asset classes over 2000–2026, models that pass the unconditional backtest breach two to three times the target rate in stress at the 95% level and five to nine times at the 99% level, and machine-learning models can be the most fragile of all. The protocol offers supervisors and practitioners a concrete instrument for the safe adoption of AI in financial risk measurement, and a foundation for subsequent work on explainability for governance and the full model-risk lifecycle.

Reproducibility. All figures and tables are generated from daily 2000–2026 public data by a single pipeline. The manuscript specifies the estimators, feature set, evaluation protocol, and configuration required to reproduce the reported analysis (§5 and Appendices B–C); the accompanying repository (Neupane, 2026) contains the data-preprocessing, estimation, forecast, and scoring code together with fixed random seeds.

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Declarations of Interest

The author reports no conflicts of interest. This research was conducted independently, using only public data and the author's own resources, and reflects the author's views alone. The author is solely responsible for the content and writing of the paper.

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code and reference list. The author designed the study, made all analytical and modelling choices, verified all reported results, and takes full responsibility for the content.

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A Empirical Evaluation Protocol

Daily data, 2000–2026, with asset-specific start dates (equities and FX from 2000, Treasuries from 2002, Bitcoin from 2014). Rolling estimation window $W = 500$ trading days; monthly re-estimation. VaR levels $\alpha \in \{0.05, 0.01\}$. Regimes are defined point-in-time: a day is labelled *stress* if its trailing 21-day realized volatility exceeds the expanding 80th percentile (minimum 252 observations), else *calm*; an exogenous crisis calendar (2008, 2011, 2020, 2022) serves as a cross-check. Coverage is tested by Kupiec LR_{uc} and Christoffersen LR_{cc}; the regime effect is tested by a logistic regression of the exceedance indicator on a stress dummy with HC1-robust standard errors. Overfitting is assessed by the probability of backtest overfitting over a five-member gradient-boosting configuration family. Calibration uses a 250-day split-conformal recalibration. No post-out-of-sample tuning is used.

B Reproducibility Details

Features: five lagged returns; lagged absolute return; 5- and 21-day trailing realized volatility. **Estimators:** historical simulation (empirical quantile); GARCH(1,1)-t (constant mean, Student-t innovations), refit monthly; gradient-boosted quantile regression (100 trees, max depth 2, learning rate 0.05, subsample 0.9, pinball loss), refit monthly; linear quantile regression on the same features, refit monthly. **Walk-forward:** expanding test index beginning after W observations; one-step-ahead forecasts only. **Scoring:** Bernoulli exceedance tests, regime-conditional breach rates, pinball loss, PBO via combinatorially symmetric cross-validation, and split-conformal coverage. All randomness is seeded; the repository (Neupane, 2026) reproduces every table and figure from the public data.